

AI assisted migration research – SonarQube report

Scope:

All six cells (methodology M1–M3 × stack Next.js / Blazor Server), scanned after generation and before any remediation. Maintainability issues were counted but not reviewed. Reliability and security issues are enumerated below with SonarQube's native category tags.

Methodology axis:

- M1: No planning.
 - Developer issues a single one-shot prompt in **agent** mode.
- M2: Autonomous planning.
 - Developer issues a single prompt in **plan** mode, that is similar in specification to the prompt used in M1.
- M3: Directed planning.
 - Developer issues a single prompt in **plan** mode, instructing the agent *how* to plan.

Source data:

SonarQube scans per cell; raw counts and issue text as observed. Time values sourced from per-cell conversation logs.

Reading note:

Distinct issues (unique rule violations) are the headline count throughout, since a rule repeated N times is one finding, not N. Total occurrences including repeats are shown alongside for recoverability against the raw scans.

1. SonarQube category tags

Every enumerated issue below carries one of SonarQube's native category tags. In this report they serve as the organizing lens:

- **Responsibility**: a security or ownership call the AI made silently (e.g. embedding credentials in source).
- **Consistency**: a non-idiomatic form used where an idiomatic one exists in the target language/framework (e.g. `isNaN` vs `Number.isNaN`, `String#replace` vs `String#replaceAll` for global replacement, redundant `alt` phrasing).
- **Intentionality**: code that will misbehave or degrade under real use (super-linear regex, non-awaited async entry point).

2. Per-cell detail

2.1 m1/s1-nextjs

Timing:

Total time: 13:20.45

- *Planning: N/A*
- *Generation: 13:20.45*

Issues:

- **1 security issue:**
 - Hard-coded database password **[responsibility]**
- **3 distinct reliability issues** (7 occurrences total):
 - Prefer `Number.isNaN` over `isNaN` — ×5 **[consistency]**
 - Redundant `alt` attribute (screen-readers already announce `img` as an image; don't use "image / photo / picture" in `alt`) **[consistency]**
 - Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking **[intentionality]**
- **88 maintainability issues** (not reviewed)

2.2 m2/s1-nextjs

Timing:

Total time: 11:53.03

- *Planning:* 02:26.75
- *Generation:* 09:26.28

Issues:

- **0 security issues**
- **2 distinct reliability issues** (6 occurrences total):
 - Prefer `Number.parseFloat` over `parseFloat` *[consistency]*
 - Prefer `String#replaceAll()` over `String#replace()` — ×5 *[intentionality]*
- **64 maintainability issues** (not reviewed)

2.3 m3/s1-nextjs

Timing:

Total time: 12:04.25

- *Planning:* 03:06.23
- *Generation:* 08:58.02

Issues:

- **0 security issues**
- **3 distinct reliability issues** (all distinct, 3 occurrences total):
 - Redundant `alt` attribute — ×2 *[consistency]*
 - Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking *[intentionality]*
- **104 maintainability issues** (not reviewed)

2.4 m1/s2-blazor

Timing:

Total time: 09:55.99

- *Planning:* N/A
- *Generation:* 09:55.99

Issues:

- **1 security issue:**
 - Hard-coded database password *[responsibility]*
- **1 reliability issue:**
 - `await RunAsync` instead. (in `Program.cs`) *[intentionality]*
- **12 maintainability issues** (not reviewed)

2.5 m2/s2-blazor

Timing:

Total time: 11:02.06

- *Planning:* 01:32.77
- *Generation:* 09:29.29

Issues:

- **0 security issues**
- **1 reliability issue:**
 - `await RunAsync` instead. (in `Program.cs`) *[intentionality]*
- **22 maintainability issues** (not reviewed)

2.6 m3/s2-blazor

Timing:

Total time: 08:13.62

- *Planning: 02:16.27*
- *Generation: 05:57.34*

Issues:

- **1 security issue:**
 - Hard-coded database password *[responsibility]*
- **1 reliability issue:**
 - `await RunAsync` instead. (in `Program.cs`) *[intentionality]*
- **8 maintainability issues** (not reviewed)

3. Cross-cell counts by severity

Distinct issues per cell. Total occurrences (including repeats) shown in parentheses where they differ.

Cell	Security	Reliability	Maintainability (not reviewed)
M1 – Next.js	1	3	88
M1 – Blazor	1	1	12
M2 – Next.js	0	2	64
M2 – Blazor	0	1	22
M3 – Next.js	0	3	104
M3 – Blazor	1	1	8

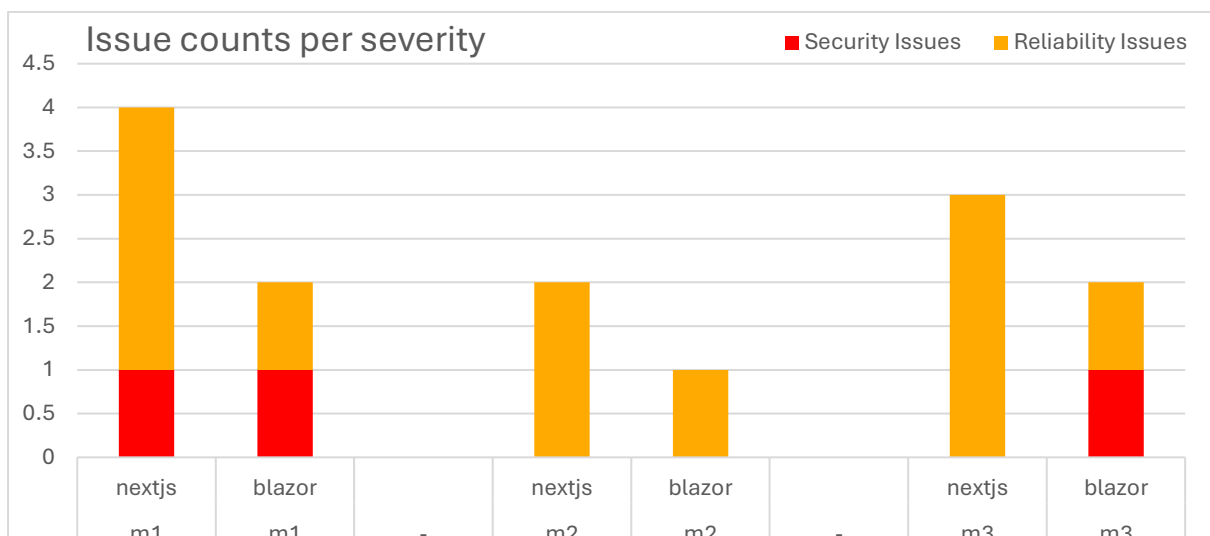


Figure1: Issue counts per severity, omitting maintainability

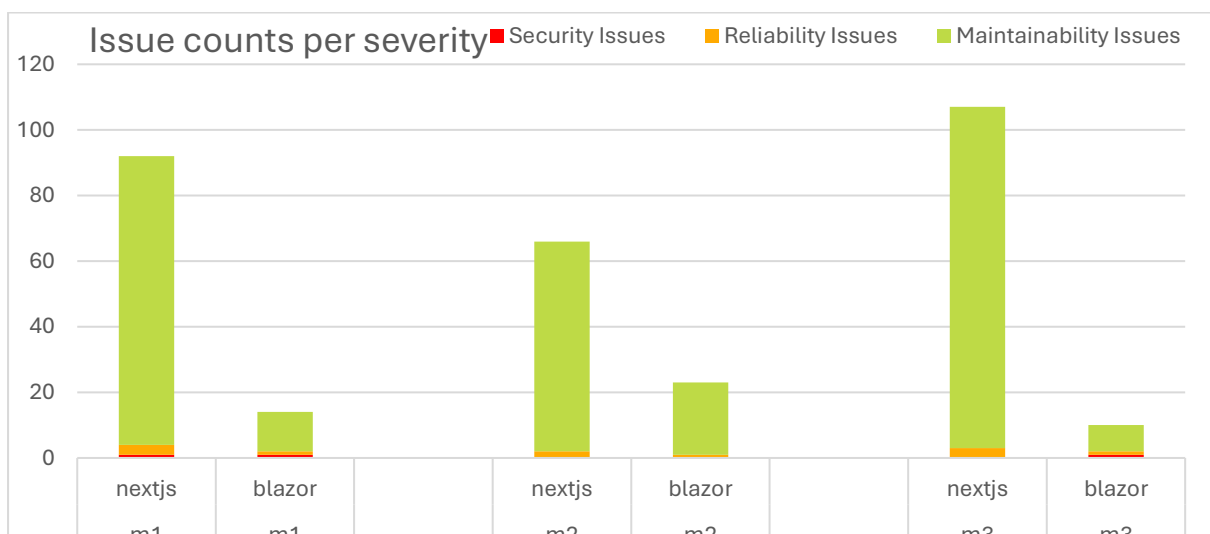


Figure2: Issue counts per severity, including maintainability

4. Cross-cell counts by category

Distinct issues, categorized per §1. Maintainability is untriaged and not tagged; totals in §3 hold the raw figure.

Cell	Responsibility	Consistency	Intentionality
M1 – Next.js	1 (password)	2 (isNaN, alt)	1 (regex)
M1 – Blazor	1 (password)	0	1 (RunAsync)
M2 – Next.js	0	1 (parseFloat)	1 (replaceAll)
M2 – Blazor	0	0	1 (RunAsync)
M3 – Next.js	0	1 (alt)	1 (regex)
M3 – Blazor	1 (password)	0	1 (RunAsync)

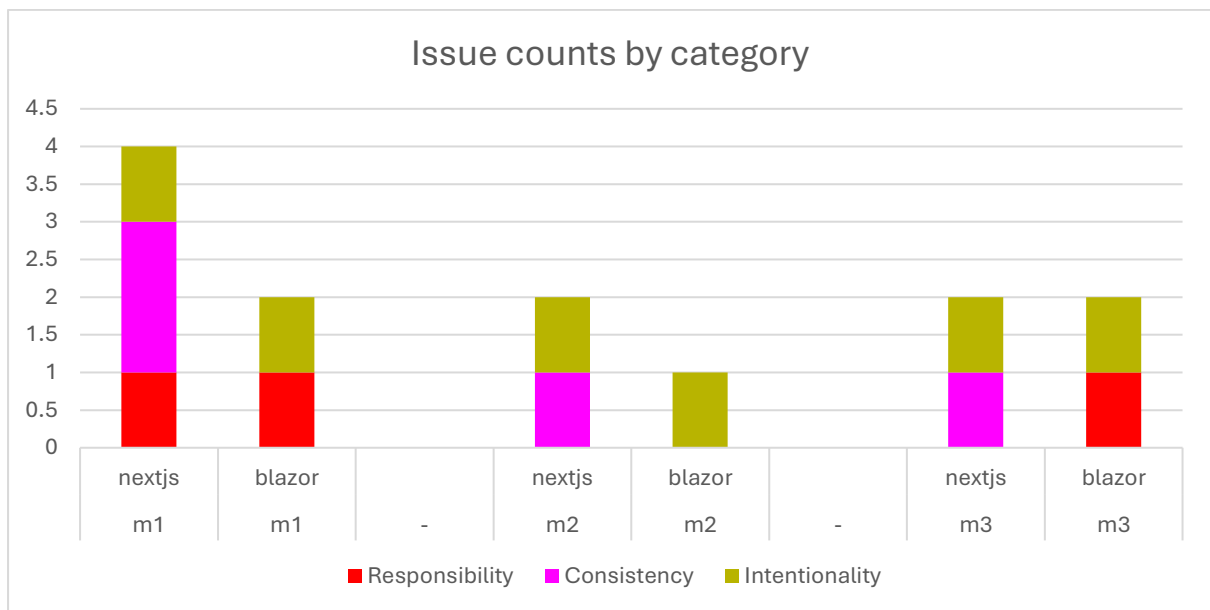


Figure 3: Issue counts by category (maintainability issues not included)